Locomotive Running Repairs

X—Setting and Repairing Flues.

In the running repairs of all locomotives the original setting of the flues cannot be gone about with too much care, especially with a view to note that all scale and sediment are thoroughly removed, both from the inside of the flue sheet as well as from the flue holes. It should also be observed that all flue holes are perfectly round and free from angular indentations, and all ragged edges should be removed, both from the inside and outside of the flue holes. The tendency to slightly injure the flue holes while removing the worn flues is very great, more especially where the older method, still in vogue in many shops, of cutting out the flues with chisels is still practised. The abrasions or indentations made by chisels, carelessly handled, are a sure means of paving the way for future leaks that are almost impossible of remedy. A chisel mark scarcely visible to the eye or touch increases in dimensions by the action of the impurities in the water, and the trouble grows with a provokingly mysterious persistence.

A fillet, which need not be large, but which should be as nearly symmetrically perfect as possible, should be made on the inside as well as on the outside of the flue holes. Where copper ferrules are used they should be annealed. Hammering hardens copper and greatly reduces its expansibility. The ends of the flues should not only be carefully swedged, but they should be thoroughly cleaned, as in rolling and expanding the flues, if the scale is left on the outer side of the flue, the process of expanding the flue has the effect of cracking the thin scale and the cracked or separated broken particles of scale, after indenting the flue sheet, gradually crumble into crystilline particles and open the way for the leaks that are sure to follow. The flues and ferrules should be a neat fit, and the copper should be of a substantial kind, that known as 40 lbs. being preferable to the lighter materials. It is good practice also to have the ferrules so constructed in point of length that they will project into the boiler not less than a quarter of an inch and not more than half an inch inside the inner edge of the flue sheet, the longer distance being preferable for the reason that scale or sediment will not adhere to copper, the projecting copper thus acting as a partial safeguard both to the flue and flue sheet.

The copper ferrule should be set in flush with the outer edge of the flue sheet, and the flue should project from three-sixteenths to a quarter of an inch outside of the flue sheet. The common rule is that the amount projecting should be at least one-and-a-half times the thickness of the flue. This will leave sufficient metal to form the bead. When the flues are clinched and held in position, a mandrel may be used to set them out firmly to the copper. The expander may then be driven into each flue and further tightened into place. The standard belling tool should then be used to turn the bead. This will leave the end of the tube turned back towards the flue sheet at an angle of about 30 degs., and the projection should be beaded from this particular position and not driven backwards in a promiscuous manner with a hammer. It will be noted that hammering the ends of the flues, unless skilfully done, has the effect of producing small cracks in the projections, which readily catch and retain small particles of burning coal and gradually extend the fracture and open the way for the eventual leak.

The rolling out of the flues should then be begun at the two upper corners. The centre flues should then be rolled, and then the remaining bottom flues. In this way the pressure is equally distributed, and is preferable to beginning at the right or left side, which may create unequal strains on the flue sheet and adjoining stay bolts. The beading should then be accomplished by the standard beading tool. It should be borne in mind that in expanding flues, the sole purpose is to tighten the tube in the hole in the flue sheet. Expanding if persisted in will stretch the opening and weaken the limited space between the flues. The experienced mechanic will readily know when the blows on the tapering pin of the expander are sounding against solid metal. The hammering and rolling should then cease. The same remarks apply to the use of an expander.

Some of the leading boiler makers approve of testing the boiler before the beading of the flues is proceeded with. The boiler is filled with warm water in the usual way, and a pressure of at least twenty-five per cent. above the working pressure is applied. This affords an opportunity of a general test of the boiler, as well as a special test of the flues. It is good practice to roll the flues tightly again after the beading process is completed, as the beading of the flues has a tendency to loosen the inner bearings of some of the flues.

The general practice in the front flue sheet is to bell out only a scattering portion of the flues, more for the purpose of partially bracing the boiler longitudinally than preventing the tendency to burn off. The heat in the front end is comparatively low, but while there is no danger in burning off the ends of the flues, it should be remembered that in some cases of boiler explosions the flues have been entirely pulled out of the front flue sheet, whereas if the flues had been properly belled out and beaded the rupture might have been averted.

In smaller running repairs, except in cases of emergency, all work on flues should be done when the boiler has had time to cool, and after being washed out. All loose and leaking tubes should be tightened by the use of the expander, and the beads properly reset. The caulking of flues is a mere temporary makeshift. It can be readily understood that a leak cannot occur in the bead of the flue. The weakness is deeper and can only be thoroughly tightened by the use of the expander. At the same time the expander must be used with much care on flues that are old and have been frequently expanded. The frequent use of the beading tool has also its dangers, as the hardened metal becomes more brittle by repeated pressures and portions of the bead will break off, exposing the end of the flue, and so hastening the decay of the joint. It is also important that the flues should be kept tight. A few leaky flues have a most pernicious effect on the flues over which the water may pass. Not only does it hasten the coming of other leaks, but it also hastens the crystallizing process to which all metals are subjected by the action of fire and water on the same surface. Wood fires are occasionally resorted to as a means for drying up small leaks. This make-believe practice should be abandoned altogether.

It need hardly be said that the careful use of tools is always necessary. The improper use of the beading tool will readily destroy a flue sheet by cutting into the metal, and the excessive use of the expander will rend the flue longitudinally and also induce the beginning of cracks in the flue sheet, especially in the limited spaces between the flues. Care should also be taken in opening any flues that may be closed. In examining the flues the hanging of a torch in the smokestack is a simple scheme for inducing a draught through the flues. A light in the firebox will readily show the flues that are closed. In the open flues the flame of the torch will readily turn into the flue, the closed flues will not draw in the flame. Roundhouses that are furnished with compressed

A leak through the quick-action parts would allow brake pipe pressure to escape at the exhaust port until the brake was applied; then if the reduction was continued until the point of equalization between brake pipe and brake cylinder pressures has been passed, the flow of air would be reversed, main reservoir pressure passing through the brake cylinders past a defective gasket, or past a defective cylinder check valve if the equalizing valve was at full stroke, would enter the brake pipe.

Leakage past the check valve under ordinary conditions can get no further than the seat of the emergency valve until brake pipe pressure has been reduced to a figure lower than pressure chamber air, plus the tension of the graduating spring and the friction of the slide valves, at which time the quick-action port would be opened and the leakage would enter the brake pipe.

The leakage through the cylinder cap gasket, however, would permit a constantly maintained brake cylinder pressure to enter the brake pipe as soon as brake pipe presssure fell lower than brake cylinder pressure.

With a neatly fitted brake valve packing ring, this leak into the brake pipe would unseat the equalizing discharge valve and cause a blow at the brake pipe exhaust port, and if the ring leaked slightly so that the equalizing reservoir could be charged as fast as the brake pipe was being supplied, there would be no blow at the brake valve exhaust, but instead an increase of brake pipe pressure, that might release some of the brakes in the train, and whether this would do so would depend upon the engine brake cylinder pressure developed, the volume of leakage into the brake pipe, the volume of leakage from the brake pipe, the volume contained in the brake pipe, the amount of brake pipe reduction, the length of time the brakes are applied, and the condition of the car brakes themselves.

For instance, if a locomotive brake cylinder pressure of 68 or 75 lbs. was being maintained after an emergency application that had developed but 58 or 60 lbs. car brake cylinder pressure, the leak from the brake cylinders would immediately start building up brake pipe pressure, releasing brakes at a time they were most needed, provided the equalizing piston packing ring in the brake valve was not a perfectly airtight fit, as mentioned before.

If a locomotive brake cylinder pressure of 50 lbs. was developed during a full service reduction and brake pipe pressure should be drawn down to, say, 45 lbs., during this time, there is no doubt but that, owing to car brake cylinder leakage, many of the auxiliary reservoirs would contain but about 45 lbs. shortly after the reduction, or be equal with brake pipe pressure, and a leak from the locomotive brake cylinder pressure of 50 lbs. into the brake pipe would immediately build up brake pipe pressure, releasing the brakes on which auxiliary reservoir pressure had reduced, but, as in the former case, it is assumed that brake pipe pressure is increased instead of the brake valve equalizing piston being unseated, and it is needless to say that if brake pipe leakage was equal to or in excess of the amount of leakage into the brake pipe. the pressure could not be increased, regardless of any other conditions.

While calling attention to this effect, it may be well to repeat that if the cause is first removed there can be no effect; therefore the importance of paying attention to the small matters in due season.

In this particular instance it will be noted that the leak into the brake cylinders will escape at the distributing valve exhaust port when from the brake pipe, but when the check valve leaks into the brake pipe it cannot be detected by the action of the brake until after the effect; therefore it is evident that a test for a leaky check valve should be conducted occasionally, and one that will answer for the purpose is, with the pressure pumped up and all other parts of the equipment in good condition withdraw brake pipe pressure with the automatic brake valve in service position; if all the pressure cannot be withdrawn, the handle should be placed on lap position and an angle cock opened, and the brake valve, being known to be free from leakage, the amount of air escaping from the hose attached to the open angle cock represents back leakage from the check valve.

Should there be no leakage from the open hose coupling, or if the brake pipe pressure can be entirely withdrawn with the brake valve, the check valve can be considered tight.

If while reducing brake pipe pressure from 50 lbs. to zero it is observed that brake cylinder pressure is falling, which would be shown by the cylinder gauge, it would indicate that pressure chamber and consequently application cylinder air is leaking past the equalizing valve packing ring and cylinder cap gasket into the brake pipe, it being understood that the application cylinder cover gasket and application cylinder pipes are tight.

Shown Up by the Test Rack.

An occurrence that will emphasize the necessity of using the triple valve test rack has recently come to our notice. A triple valve has been removed from a freight car in order to be cleaned and tested, as a year had elapsed since the triple valve had been cleaned, and when being tested it was found that the valve would not pass the service port capacity test. In an effort to open the service port the slide valve bushing was forced out of the body, and it was found that there was no service port in the casting. In fact, there never had been one in that body. Dates stamped on the flange of the body showed it to have been in service in the year of 1907.

No doubt this triple valve frequently worked in the service applications with other triple valves under favorable conditions by expanding auxiliary reservoir pressure through the emergency port and past the loosely fitted emergency piston, but it is safe to say that it did not do this under all conditions, and there is no doubt that this valve was responsible for many a case of "dynamiting" or train parting.

There are many defects of triple valves that cannot be detected unless by the use of the test rack, and this instance should not fail to convince the most skeptical repairman of the old school of the absolute necessity for the use of triple valve test racks.

Impossibilities that George Westinghouse Has Overcome.

George Westinghouse is the greatest inventor of this age. He is always inventing something that would make any other man famous, but he keeps on in a modest way without boasting of what was coming, and everything he produces is a success. Commenting on some Westinghouse inventions, the New York *Tribune* says:

"George Westinghouse has been overcoming obstacles all his life. During the last three years there have been more obstacles than usual. But, as Andrew Carnegie is reported to have said: "George Westinghouse is a genius; you can't keep him down."

"Forty-two years ago they said it was 'impossible' to stop a railroad train with air. But he did it.

"Force natural gas through hundreds of miles of pipe, regulate its flow, safely supply the homes and aid the industries of Pittsburgh, with cheap and invisible fuel? 'Impossible!' But he did it.

"Build great gas engines that would work as smoothly as watches and could be connected direct to the dynamo shafts? 'Impossible!' But he did it.

"Use the alternating current for transmitting electricity over long distances, supplying light and power to homes, halls, stores, factories, railroad trains? 'Impossible!' But he did it.

"Build dynamos that would harness Niagara? 'Impossible!' But he did it. And several other 'impossible' things. Probably that is why he never seems to be disturbed when anybody tells him that this or that is 'impossible' to accomplish by engineering.

"There at the works the men know that he can do what he starts out to do. Even years ago they had this saying: 'When the boss is on the job all we have to do is to hand him the tools.'"

engine broke a valve stem so that one side of the engine was inoperative, the engineer would have no hesitation in trying to handle the train with his engine on one side, and no doubt he could handle it successfully. In this case, if the engine was disconnected in the old manner—that is, both side rods taken down—the probabilities are that it would be necessary to send for another engine to handle the train. For this reason, would it not be better to disconnect the main rod on the same side where the side rod was broken and leave the side rod up on the other side, thus virtually putting the engine on one side and thereby leaving it its total ratio of adhesion so as to eliminate all possibility of slipping?

Considering Law No. 2, "failure of main connection," there are some types of engines, notably switch engines, which have the main rod connected to the rear pair of drivers and the eccentrics on the intermediate pair. In case of failure of a main connection of an engine of this type, if the corresponding side rod on the opposite side were taken down also, the engine would be totally disabled. This also applies to some types of consolidation engines where the main rod is connected to one pair of wheels, while the eccentrics are mounted on the axle of the preceding pair, and in this case, same as the one above mentioned, in case of the failure of the main connection on one side it would be necessary to take down all side rods on both sides according to the old methods of disconnecting, which would totally disable this engine just the same as the previous one. While there cannot be such serious objection to the old methods of disconnecting in case of an 8-wheel engine, yet we believe that in modern practice different methods should be employed, as, for instance, in case of the failure of the main connection on any type of engine where the main rods are connected to one pair of wheels and the eccentrics are mounted on another axle, instead of totally disabling the engine by disconnecting all side rods, the side rods should be removed only from the side where the main connection is broken, but, in addition to this, the main rod should be taken down on that side also. This would put the engine on one side, and not only enable it to handle itself, but to handle about one-half of its usual rating.

Again, take the case of a broken main connection on an ordinary mogul or 10-wheel or consolidation engine, where the eccentrics are mounted on the same axle to which the main rods are connected. In this case, if the main connection on one side broke, established custom would have you remove all side rods from both sides, bringing the engine in with the main drivers only. If this were on a consolidation engine, we question whether the engine would handle itself with but the main drivers operative, as in this case the ratio of adhesion to tractive power would be as 1 to 1, and, consequetntly, the main wheel would simply slip instead of propelling the engine. A 10-wheel engine disconnected in this manner would necessarily have to give up its train, whereas, if on a passenger train and disconnected by the new method, it could handle its train to the terminal and possibly make running time.

Considering Law No. 3, "in case of a broken main crank pin close to the wheel," this, under the old method, would mean take down all side rods on both sides and bring the engine in with one main rod up only. In case of anything larger than an 8-wheel engine it is question whether the engine would handle itself or not, and even if it did handle itself, which would only be possible on a level road, it could not handle any portion of its train. In our opinion, therefore, the proper method of disconnecting would be to take down the main rod and all side rods on the disabled side, leaving the main rod and all side rods up on the good side. This would not only enable the engine to handle itself, but if on a passenger train to handle the passenger train, and if on a freight train to handle about one-half of its rating. Another point in favor of the new method of disconnecting is that on nearly all modern engines larger than the 8-wheel type it is necessary to disconnect the back end of the main rod before the side rods can be taken down; therefore, under the old method, after the side rods were taken down, it would be necessary to connect the back ends of the main rod up again. This would involve a possibility of getting liners mixed up, thereby lengthening or shortening the main rod so that in addition to the broken side rod or crank pin, as the case might be, you would probably have a broken cylinder head, if not a broken cylinder.

We fully realize that objections will be heard from all sides against the proposed method of disconnecting, some of the objections being that with one main rod up only and all the side rods up on the same side there would be a possibility of some of the drivers slipping, so as to throw the rods in strain, breaking either the rods or the crank pins. A moment's thought, however, should convince anyone that the wheels will not slip going down hill when the engine is drifting, and cannot slip when working steam, for the simple reason that, as stated before, the ratio of adhesion to tractive power is usually as 4 or 4.5 is to 1 when both main rods of the engine are connected. With only one side of the engine up the ratio of adhesion would be more than double; consequently, the one cylinder would not develop enough power under any circumstances to slip the drivers. Cases will no doubt be cited where damage was done by leaving the side rods up on one side and disconnecting same on the other. A careful investigation of these cases would no doubt develop the fact, however, that in all such instances either both main rods were left up or else the main rod was left up on one side and the side rods on the other. In view of the ever-increasing dimensions of locomotives, with a corresponding increased difficulty in handling heavy rods, we can see no reason why we should not depart from the old-time methods and adopt methods more in line with modern railroading. The subject is now open for discussion.

Inventions That Are Not Pushed.

When any new invention or process has been brought out people in the regions of modern civilization strive to improve it in every way they can think of. The various peoples from whom we inherited the germs civilization were deeper steeped in a spirit of contentment than we are, and they did not harass themselves to better things that served their turn in a modest manner.

Take steel making, for instance. The natives of India made Wootz steel two thousand years before Huntsman, the reputed inventor of cast steel, was born. The Indian blacksmiths made their cast steel in small lumps by a very crude process, but their product was the steel from which the famous Damascus swords were made; so its quality was seldom excelled. These Indian blacksmiths are making their steel by the old process today, but the output is so limited that railway companies in Hindustan cannot get enough native steel to supply their machine shops with cutting tools.

Then take butter making. This useful article of diet was first made in Syria and was discovered by accident. A sheik noted for the number of his flocks and herds preferred milk to the juice of the grape as a beverage, and he had a supply of milk carried in goat-skins. The jolting of the camels churned the milk and the resulting thick substance was at first thrown away, but by degrees the people found that the thick stuff had merits all its own, and butter had come into use. The Syrians—Arabs—have not improved on that process of making butter, which in those parts continues to be more of a liquid than a solid.

Most of our readers are no doubt familiar with Lamb's story of how the Chinese learned to cook roast pig. But that is a ridiculous story, illustrating habits of primitive people.

consequently the water must be lifted the amount of difference between the surface of the water and the injector opening. In order to lift the water, a vacuum is produced in the pipe leading from the tank to the injector, by admitting steam into the injector and allowing it to pass through into the overflow pipe. The interior of the injector is so constructed that the steam finds its way into the overflow pipe much more readily than it does into the pipe leading to the tank. The air in the pipe leading to the tank is induced to mix with the steam and pass out at the overflow pipe, while the weight of the atmosphere on the surface of the water in the tank forces the water into the vacuum thus produced. The joints forming the connection between the injector and the tank should be of the most secure kind, as a leak in this pipe will greatly diminish, if it does not altogether stop the operation of the injector. In some experiments where injectors are placed below the level of the water in the tank the improvement in the working of the injector is of a marked kind.

In all injector troubles the failure of the water to reach the injector is the most common, and if there is water in the tank and the surface be not frozen, as it may readily be in winter, the trouble will likely be by reason of a leak in the suction pipe or by clogging of the strainers. A heated suction pipe also often refuses to supply water in sufficient quantity for the effective use of the injector. In cases where the injector lifts the water and fails to force it into the boiler, the trouble should be readily evident by the behavior of the injector. If the steam should blow back into the tank it will either be caused by reason of the overflow valve being partly closed, thereby preventing a ready escape for the combined steam and water which is necessary until the mixture of the two bodies have accumulated sufficient momentum to pass through the check valve opening into the boiler, or it may be caused by the sticking of the check valve which, though rarely the case, sometimes happens in instances where impurities in the water tend to form solutions that are particularly adhesive in their nature, acting like glue in joining the check valve to the valve seat. A slight tap on the check will sometimes relieve a sticking valve.

Some simple remedies there are that may be temporarily effective as in the case of clogged strainers or obstructions in the suction pipe. In such cases the overflow valve should be closed and the steam blown back through the suction pipe into the tank. This may clear the pipe but the tank and the strainer should be cleaned as speedily as possible, as the tendency to accumulate impurities in the tank is very great, and nearly all strainers in use in locomotive tanks not only collect impurities but hold them at the entrance of the suction pipe where detached particles sooner or later find their way through and so increase the chances of injector troubles.

As is well known, the degree of perfection to which the mechanism of the injector has attained has been the result of very extensive and careful experiments, the taper openings compressing the mixture of steam and water faciliating and reducing the current in one direction and retarding or entirely checking it in another, are not subject to improvement by changes in their relation to each other, therefore the repairing of injectors should be intrusted only to the most proficient mechanics—specialists familiar with the parts. Specialists do not spring like Athene fully armed from the brow of Jove, but acquire their knowledge by long and careful practical experience. Parts furnished by the original constructors are usually nearer perfection in detail than rough and ready substitutes, and the trained mechanic will prefer their use when a replacing of worn parts is necessary. Much may be saved by a systematic cleaning of injectors, especially where deposits of carbonate of lime are formed. Injectors may be readily cleared of all incrustations by immersing them in a bath of benzine or diluted muriatic acid, the mixture being ten or twelve parts of water to one of acid.

In taking an injector apart it will often be found that the joints have acquired a degree of tightness which almost renders them liable to distortion by reason of the force required to loosen the joints. In such cases it will be found that slightly heating the joints aids greatly in their liability to slacken, as brass expands rapidly in heating. In many shops a supply of injectors, cleaned, repaired and tested, are usually kept on hand, so that when an injector is reported to be defective it can be disconnected and another put in its place, thus avoiding delay and allowing the skilled mechanics to examine the injector under favorable conditions and with proper tools at hand. The practice of striking injectors with hammers and other hardened tools is a very objectionable one. The blows rarely have the effect of dislodging any obstruction in the chambers or tubes of the injectors, while the fine appearance of the injector is irretrievably destroyed.

Lubricators.

The almost universal application of lubricators to the locomotive is one of the most useful attachments ever applied to the steam engine. Like the injector, its perfection has not been the work of a day. A constant supply of oil to the running parts of an engine accurately gauged to the requirements of the service was something almost beyond mere human possibility of accomplishment. The application of steam or compressed air to the oil was an important step, culminating in the introduction of a series of force pumps supplying a positive and regular supply of oil, which had hitherto been impossible to the parts where the back-pressure of steam interfered with the free and constant flow of the lubricant. Obstructions, of course, may interfere with the flow of oil at any time, and if from any cause the appliance ceases to perform its functions, the best method is in the case of the common lubricator to open the bottom cocks and allow a full pressure of steam to blow through. If the trouble continues, the glass tubes should be removed and the small feeders carefully examined, when it will likely be found that the small openings have been choked with some substance easily removed. If the openings are perfectly clear, the top chamber should be removed, when the feed pipes may be readily observed. Their liability to break or become stopped up is not great, but it is possible, and they should be occasionally cleared and tested. The quality of the oil is also of considerable importance in the working of the lubricator, as well as in the lubrication of the machinery and it is generally discovered that the cheapest lubricant is nearly always the worst.

Questions Answered

DRIVER BRAKE RUBBING WHEEL.

62. R. K., Indianapolis, Ind., writes: How can you keep the brake shoes from rubbing continuously on the tires with cam-driver brakes? A.—The continuous rubbing of these brake shoes is generally caused by the shoe not being properly balanced as it is suspended by a pin in the hanger. A piece of heavy bent iron wire or a small bent iron rod placed between the shoe and the hanger will keep the point of the shoe away from the wheel and not in the least interfere with the application of brakes.

END PLAY OF DYNAMO SHAFT.

63. G. B. S., Three Forks, Mont., writes as follows: In a baggage car containing a dynamo for train lighting, which has a shaft placed lengthwise of the car. When the car is maintaining an even speed of forty miles an hour on level track dynamo running, will the shaft be carried in its bearings the same as when train is standing, or will it crowd the rear bearing causing hub friction? If so, how much with 3½-in. shaft, 8 ft. long, bearings 6 in. long, shaft and attached weighing 850 lbs.?—A. The shaft will rotate in its bearings just as if the train was at rest. A good proof of this is a passenger seated in a car traveling at high speed. The passenger does not feel any tendency to move forward or back. If, however, the

a train of light cars, the speed, nominal percentage of braking power, and track conditions being equal.

One reason that is not given much thought is the difference in brake cylinder pressure derived from service and emergency application.

When an 8-in. brake equipment is used an emergency application will result in a substantial gain in cylinder pressure over that derived from the service application, but as 16 and 18-in. brake cylinders are used, the brake pipe remains of the same size, and the air that enters the brake cylinder from the brake pipe is not much more than equal to the volume of air required to displace the brake piston.

Another manner in which the actual brake cylinder pressure is affected is by the increase of piston travel, due to the emergency application when cars are in motion. The losses in the percentage of braking power that occur through the foundation brake gear of heavy cars is generally understood, and air brake men have for some time realized that the brake equipments in use were unable to meet the demands upon them by modern trains.

During the Lake Shore demonstrations it was noted that 2½ seconds time elapsed between the time of brake application and brake effectiveness, even with the most modern "L. N." passenger brake. A special triple valve known as the "L. G. N." was then used in an effort to develop 105 lbs. brake cylinder pressure from a 110-lb. brake pipe pressure, and in spite of the effect of emergency piston travel, came within 2 lbs. of producing the desired result, but the lapse of time between the brake application and its effectiveness still remained at 2½ seconds; whereupon it was decided that in order to reduce this time to 2 seconds, or less if possible, the use of triple valves must be dispensed with.

In order to meet this condition, the Westinghouse Air Brake Company designed and perfected the control valve and "P. C." equipment. And by the use of this equipment the aforesaid elapsed time was reduced to two seconds, and the stop from a speed of sixty miles per hour was actually made in 1,100 ft.

The control valve used contains the features of the distributing valve, and several very important features in addition, and the use of large posts and larger pipes make possible the prompter effect of brake application.

From the general arrangement it will be observed that two brake cylinders are employed—one for service operation, both for emergencies, which means that the service braking power is doubled during emergency applications. The brake can be applied to any weight of car, and the control valve works in harmony with triple valves.

The control valve contains the principal features of the distributing valve in maintaining brake cylinder pressure against leakage and excessive piston travel, and a feature in addition that results in the application of the emergency cylinder if for any reason brake pipe pressure leaks down to a predetermined figure.

Just how the valve accomplishes this and its construction and operation will be explained in future issues.

"Brakes Creeping On."

Frequent inquiries concerning the subject of brakes creeping on inspires us to offer a few comments on this matter. When the question, "What causes the brake to creep on when the valve handles of the H6 brake are in running position?" is asked, the invariable reply is, "The same thing that causes the old style or A1 type of brake to creep on when the brake valve handle is in running position."

Namely, this is due to brake pipe leakage in combination with a feed valve that will not promptly open and supply this leakage.

The explanation is generally accepted as satisfactory, and when either of the disorders mentioned are corrected the trouble on the lone engine will disappear.

However, the reapplication of the brake on the head end of a long train after a release will also affect the engine and tender brake, especially if they are in good condition, and this sometimes confuses the engineer upon the subject of brakes creeping on.

The reapplication is sure to follow the overcharge of the brakes on the head end of the train, and this trouble of brakes creeping on the engine is generally more frequent when the H6 brake is in use.

The reason for this is that the average condition of the brake on the locomotive, when a triple valve and auxiliary reservoir are used, is such that it usually requires a rather heavy and rapid reduction of brake pipe pressure to get the brake applied when desired, and as a result the slight overcharge fails to affect the engine brake to any noticeable extent. The operation of the No. 6 distributing valve does not depend upon a fixed charge in an auxiliary reservoir of limited volume, and a movement of the equalizing valve to application position results in an application of the brake, regardless of piston travel and cylnder leakage, while the movement of the triple valve piston under similar conditions does not always result in an application of the brake. With the No. 5 distributing valve the overcharge is not followed by an application of the driver brake on account of the application chamber being open to the atmosphere while the handles are in running position regardless of the position of the equalizing valve, but the same variation in brake pipe pressure is encountered after the overcharge no matter what type of engine brake is used.

Let us attempt to consider the subject of brakes creeping on without referring to any particular type of air brakes as we will try to make it plain that every automatic air brake applies from the same cause and from that cause only, otherwise it would not be considered automatic.

The movements of this earth and the planets are governed by fixed law, the movement of every thing mechanical since the creation is governed by the same fixed law that does not vary and we do not expect to see water flow up hill or rise above its own level, when influenced by nothing other than gravitation and atmospheric pressure any more than we would expect the earth to revolve in the opposite direction tomorrow.

We will not consider any movement of the earth or planets, save as an illustration and the law governing the operation of the automatic air brake is as positive and fixed, and is no more likely to vary than the law which compels and restrains the movement of celestial bodies.

The law which underlies the operation of the automatic brake is the creation of differentials of pressure. The differential may be created without necessarily resulting in an operation of the mechanism, but the mechanism cannot be operated save by the creation of the differential in pressure and having this fixed in our mind we will assume that by brakes "creeping" on, we mean an automatic application, a movement of the triple valve, or the equalizing valve of the distributing valve, and not a stuck or sticking brake.

Very often this disorder has been reported on an engine, sometimes the report was merely "clean triple valve" or "clean distributing valve," it being taken for granted that this work if performed would correct the evil or remedy the disorder.

Even when brakes "creeping" on has been reported the first move of the repairman of the old school was to take a hammer, chisel and a monkey wrench and make an assault upon the triple valve, probably the brake valve next, and as a general result very little if anything was accomplished, and very often there was nothing wrong in the first place except an ever-charged brake pipe.

There is really nothing difficult to locate and very little to reason out in